

Test ID	Input Expression	Expected Result	Type	Algorithm A Result	Algorithm B Result	Status A	Status B	Notes
TC01	$3 + 4 * 2$	11	Valid	11	11	PASS	PASS	Operator precedence
TC02	$(3 + 4) * 2$	14	Valid	14	14	PASS	PASS	Parentheses
TC03	$((8 + 2) * 5)$	50	Valid	50	50	PASS	PASS	Nested parentheses
TC04	$(3 + 4$	Error	Invalid	Error	Error	PASS	PASS	Missing closing parenthesis
TC05	$3 + 4)$	Error	Invalid	Error	Error	PASS	PASS	Extra closing parenthesis
TC06	$3 + * 4$	Error	Invalid	Error	Error	PASS	PASS	Invalid operator sequence
TC07	$10 / (5 - 5)$	Error	Invalid	Error	Error	PASS	PASS	Division by zero
TC08	$2 + 3 * 4 - 5$	9	Valid	9	9	PASS	PASS	Multiple operators and precedence
TC09	$5 * 6 + 3$	33	Valid	33	33	PASS	PASS	Operator precedence
TC10	$100 / 5 / 2$	10	Valid	10	33	PASS	PASS	Parentheses and Operator precedence